

Gumbel-max Watermark

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Lemma 1.

If $X \sim U(0, 1)$, $Y = -\ln(X)$, then $Y \sim \text{Exp}(1)$

Proof.

This is clear by manipulating their CDF.

Now

$$\arg \max_w \frac{\log U_w}{P_w} \iff \arg \min_w \frac{-\log U_w}{P_w} \iff \arg \min_w \frac{\text{Exp}(1)}{P_w}$$

Note (by property of Exp or Gamma, as we learned in stat 450),

$$Y_w := \frac{\text{Exp}(1)}{P_w} \sim \text{Exp}(\lambda = P_w)$$

By stat 333,

Lemma 2.

Let $\{X_i\}_{i=1}^n$ be a sequence of independent rvs where $X_i \sim \text{EXP}(\lambda_i)$, $i = 1, 2, \dots, n$. For $j \in \{1, 2, \dots, n\}$, $\mathbb{P}(X_j = \min\{X_1, X_2, \dots, X_n\}) = \frac{\lambda_j}{\sum_{i=1}^n \lambda_i}$

Proof.

$$\begin{aligned}
& \mathbb{P}(X_j = \min\{X_1, X_2, \dots, X_n\}) \\
&= \mathbb{P}(X_j < X_1, X_j < X_2, \dots, X_j < X_n) \\
&= \int_0^\infty \mathbb{P}(X_j < X_1, X_j < X_2, \dots, X_j < X_n \mid X_j = x) \lambda_j e^{-\lambda_j x} dx \\
&= \int_0^\infty \mathbb{P}(X_1 > x, X_2 > x, \dots, X_n > x) \lambda_j e^{-\lambda_j x} dx \quad (\text{since } X_j \text{ is independent of } \{X_i\}_{i \neq j}) \\
&= \int_0^\infty \mathbb{P}(X_1 > x) \mathbb{P}(X_2 > x) \cdots \mathbb{P}(X_n > x) \lambda_j e^{-\lambda_j x} dx \\
&= \int_0^\infty e^{-\lambda_1 x} e^{-\lambda_2 x} \cdots e^{-\lambda_n x} \lambda_j e^{-\lambda_j x} dx \\
&= \int_0^\infty \lambda_j e^{-(\sum_{i=1}^n \lambda_i) x} dx \\
&= \frac{\lambda_j}{\sum_{i=1}^n \lambda_i} \int_0^\infty \left(\sum_{i=1}^n \lambda_i \right) e^{-(\sum_{i=1}^n \lambda_i) x} dx \\
&= \frac{\lambda_j}{\sum_{i=1}^n \lambda_i}
\end{aligned}$$

□

Therefore

$$\mathbb{P}(Y_w \text{ is the minimum}) = \frac{P_w}{1} = P_w$$

and $\{\arg \min_{k \in W} Y_k = w\} = \{Y_w \text{ is minimum}\}$ as events, so we are done.

Proof of Theorem 2.1

Consider equation (30):

- Consider the 1st inequality.

Observe

$$\mathbb{E}_1[T_h(Y_{1:n})] = 1 \cdot \mathbb{P}_1(T_h = 1) + 0 \cdot \mathbb{P}_1(T_h = 0) = \mathbb{P}_1(T_h = 1)$$

And so

$$1 - \mathbb{E}_1[T_h(Y_{1:n})] = \mathbb{P}_1(T_h = 0)$$

Write

$$\mathbb{P}_1(T_h = 0) = \mathbb{P}_1\left(\sum_{t=1}^n h(Y_t) < n\right) = \mathbb{P}_1\left(\sum_{t=1}^n -h(Y_t) > -n\right)$$

If $\sum_{i=1}^n -h(Y_t) > -n$, then $\sum_{i=1}^n -h(Y_t) > -n$, thus

$$\exp\left(\sum_{i=1}^n -h(Y_t)\right) > \exp(-n)$$

By Markov, for nonnegative, $a > 0$, $\mathbb{P}(\geq a) \leq \frac{\mathbb{E}[\cdot]}{a}$, so

$$\mathbb{P}_1 \left(\sum_{i=1}^n -h(Y_t) > -n, \right) \leq \frac{\mathbb{E}_1[\exp(\sum_{i=1}^n -h(Y_t))]}{\exp(-n)}$$

(In discrete case, note $\mathbb{P}(> a) \leq \mathbb{P}(\geq a)$. In cts case, they are equal.)

- Consider the 2nd inequality

Since

$$\mathbb{E}_{H_1} \exp \left(\sum_{t=1}^n -h(Y_t) \right) = \mathbb{E}_{H_1} \left[\prod_{t=1}^n \exp(-h(Y_t)) \right]$$

By tower rule,

$$\begin{aligned} \mathbb{E}_{H_1} \left[\prod_{t=1}^n \exp(-h(Y_t)) \right] &= \mathbb{E}_{H_1} \left[\mathbb{E}_{H_1} \left[\prod_{t=1}^n \exp(-h(Y_t)) \mid (w_{1:(t-1)}, 1:(t-1)) \right] \right] \\ &= \mathbb{E}_{H_1} \left[\left(\prod_{t=1}^{n-1} \exp(-h(Y_t)) \right) \mathbb{E}_{H_1} [\exp(-h(Y_n)) \mid (w_{1:(t-1)}, 1:(t-1))] \right] \end{aligned}$$

By **working hypothesis 2.1** (here P_t is known and t independent), this is

$$\mathbb{E}_{H_1} \left[\left(\prod_{t=1}^{n-1} \exp(-h(Y_t)) \right) \mathbb{E}_{1, P_n} [\exp(-h(Y_n))] \right]$$

Note

$$\mathbb{E}_{1, P_n} [\exp(-h(Y_n))] = \exp(\log P_{n,h}()) \leq \exp(\log p_{h}())$$

where $p_{h}() = \sup_{P \in \mathcal{P}} P_h() = \sup_{P \in \mathcal{P}} \mathbb{E}_{1, P} [\exp(-h(Y))]$.

And by repeatedly doing above from $t = n - 1$ to $t = 1$,

$$\mathbb{E}_{H_1} \left[\prod_{t=1}^n \exp(-h(Y_t)) \right] \leq (\exp(\log p_{h}()))^n = \exp(n \cdot \log p_{h}())$$

For the last equality in the proof,

$$\inf_{\geq 0} \exp \left(\mathbb{E}_0 h(Y) \log p_{h}() \right) = \exp \left(\inf_{\geq 0} [\mathbb{E}_0 h(Y) \log p_{h}()] \right)$$

as $\exp$ is continuous and monotone increasing.

Intuition for equation (29) and its proof

To control type 1 error, for $\epsilon \in (0, 1)$

$$\mathbb{P}_{H_0} \left(\sum_{t=1}^n h(Y_t) \geq n, \right) =$$

which is

$$\mathbb{P}_{H_0} \left(\frac{1}{n} \sum_{t=1}^n h(Y_t) \geq \frac{n}{n} \right) =$$

Under H_0 , Y_i are iid. Thus by SLLN,

$$\frac{1}{n} \sum_{t=1}^n h(Y_t) \xrightarrow{a.s.} \mathbb{E}_0 h(Y)$$

So we guess

$$\lim_{n \rightarrow \infty} \frac{n_i}{n} = \mathbb{E}_0 h(Y)$$

because otherwise, will go to either 0 or 1.

Using contradiction, the paper shows

$$\limsup_{n \rightarrow \infty} \frac{n_i}{n} \leq \mathbb{E}_0 h(Y)$$

and

$$\liminf_{n \rightarrow \infty} \frac{n_i}{n} \geq \mathbb{E}_0 h(Y)$$

and so the limit exists and equals $\mathbb{E}_0 h(Y)$.

KL Divergence and Donsker-Varadhan

Definition 3.

Let P and Q be two probability measures on a measurable space $(\mathcal{X}, \mathcal{F})$. Assume P is absolutely continuous with respect to Q (denoted as $P \ll Q$), meaning that for any measurable set A , if $Q(A) = 0$, then $P(A) = 0$. By the Radon-Nikodym theorem, there exists a density function (or likelihood ratio) $\frac{dP}{dQ}$ such that $P(A) = \int_A \frac{dP}{dQ} dQ$.

The KL divergence of P from Q is defined as:

$$D_{\text{KL}}(P||Q) = \mathbb{E}_P \left[\log \frac{dP}{dQ} \right] = \int_{\mathcal{X}} \log \left(\frac{dP}{dQ} \right) dP$$

If P is not absolutely continuous with respect to Q , then $D_{\text{KL}}(P||Q) = \infty$.

Remark 4.

Note that every distribution is a probability measure (push forward measure):

$F_X(x) = \mathbb{P} X^{-1}((-\infty, x])$. By Carathéodory extension thm, CDF uniquely determines its probability measure.

Remark 5.

By expectation Jensen's inequality, where $-\log$ is convex, $D_{\text{KL}}(P\|Q) \geq 0$, and $D_{\text{KL}}(P\|Q) = 0$ if and only if $P = Q$ almost everywhere.

Theorem 6 (Jensen's inequality).

Suppose that X is a random variable with expectation μ , and function g is convex and finite. Then

$$\mathbb{E}_X[g(X)] \geq g(\mathbb{E}_X[X])$$

with equality if and only if, for every line $ax + b$ that is a tangent to g at μ

$$\mathbb{P}_X[g(X) = ax + b] = 1.$$

<https://www.math.mcgill.ca/dstephens/556-2014/Handouts/Math556-05-Inequalities.pdf>

Corollary 7.

Suppose g is strictly convex and finite. Then $\mathbb{E}[g(X)] = g(\mathbb{E}[X])$ if and only if $\mathbb{P}[X = \mu] = 1$

Proof.

$\Rightarrow$: Assume $\mathbb{E}[g(X)] = g(\mathbb{E}[X])$. By theorem, for any tangent line $(x) = ax + b$ to g at the point μ , we have $\mathbb{P}[g(X) = ax + b] = 1$. Because g is convex and finite, it possesses at least one subgradient at μ (This is fact 9.12 and corollary 9.13 in my co463 notes). Therefore, there exists at least one line $(x) = ax + b$ that is tangent to g at μ . This guarantees the existence of a and b such that $g(\mu) = a\mu + b$. By the definition of strict convexity, a strictly convex function intersects any of its tangent lines at exactly one point: the point of tangency. Therefore, for all $x \neq \mu$, we have a strict inequality: $g(x) > ax + b$. Consequently, the equation $g(x) = ax + b$ holds true if and only if $x = \mu$. Define the events $A = \{\omega \in \Omega \mid g(X(\omega)) = aX(\omega) + b\}$ and $B = \{\omega \in \Omega \mid X(\omega) = \mu\}$. Then $g(X(\omega)) = aX(\omega) + b \iff X(\omega) = \mu$ holds for every ω by above. That is $A = B$. Recall we know $\mathbb{P}(A) = 1$. So $\mathbb{P}(B) = 1$, which means $\mathbb{P}[X = \mu] = 1$.

$\Leftarrow$: Assume $\mathbb{P}[X = \mu] = 1$. Let $(x) = ax + b$ be an arbitrary tangent line to g at μ . 4. Because (x) is tangent to g at μ , then $g(\mu) = a\mu + b$. Since $\mathbb{P}[X = \mu] = 1$, then $\mathbb{P}[g(X) = g(\mu)] = 1$, $\mathbb{P}[aX + b = a\mu + b] = 1$. Since $g(\mu) = a\mu + b$, then $\mathbb{P}[g(X) = aX + b] = 1$. This holds for every line $ax + b$ tangent to g at μ . Thus by theorem, $\mathbb{E}[g(X)] = g(\mathbb{E}[X])$. □

Theorem 8 (The Donsker-Varadhan Representation).

For any two probability measures P and Q on $\mathcal{X}$,

$$D_{\text{KL}}(P\|Q) = \sup_f (\mathbb{E}_P[f] - \log \mathbb{E}_Q[e^f])$$

where the supremum is taken over measurable extended-real-valued functions $f : \mathcal{X} \rightarrow [-\infty, \infty)$ such that f is integrable under P and $0 < \mathbb{E}_Q[e^f] < \infty$.

We use the convention $e^{-\infty} = 0$

Remark 9.

We say "taken over any measurable function such that the expectations are well-defined" is because, if we only require boundedness (which is the bare minimum if we want the representation to be well defined), the optimal function $f^* = \log\left(\frac{dP}{dQ}\right) \log c$ below is usually not bounded.

Proof.

We prove it by proving

- $\mathbb{E}_P[f] - \log \mathbb{E}_Q[e^f] \leq D_{\text{KL}}(P\|Q)$ for all f ,
- the equality is achieved for some specific f .

Assume $D_{\text{KL}}(P\|Q) < \infty$, and so $P \ll Q$. Let $p = \frac{dP}{dQ}$ be the Radon-Nikodym derivative.

Consider the quantity $\mathbb{E}_P[f] - D_{\text{KL}}(P\|Q)$. We can write this as a single integral with respect to P :

$$\mathbb{E}_P[f] - D_{\text{KL}}(P\|Q) = \int_{\mathcal{X}} f dP - \int_{\mathcal{X}} \log(p) dP = \int_{\mathcal{X}} \log(e^f) dP - \int_{\mathcal{X}} \log(p) dP = \int_{\mathcal{X}} \log\left(\frac{e^f}{p}\right) dP$$

Since $\log$ is concave, by Jensen's inequality, $(\mathbb{E}[\log(\cdot)] \leq \log(\mathbb{E}[\cdot]))$:

$$\int_{\mathcal{X}} \log\left(\frac{e^f}{p}\right) dP \leq \log\left(\int_{\mathcal{X}} \frac{e^f}{p} dP\right)$$

By Radon-Nikodym, $dP = p dQ$,

Let $\mathcal{X} = \{p > 0\}$, then

$$\int_{\mathcal{X}} \frac{e^f}{p} dP = \int_{\mathcal{X}} \frac{e^f}{p} dP = \int_{\mathcal{X}} e^f dQ \leq \int_{\mathcal{X}} e^f dQ$$

and note

$$\log\left(\int_{\mathcal{X}} e^f dQ\right) = \log \mathbb{E}_Q[e^f]$$

Thus

$$\mathbb{E}_P[f] - D_{\text{KL}}(P\|Q) \leq \log \mathbb{E}_Q[e^f]$$

Which is

$$\mathbb{E}_P[f] - \log \mathbb{E}_Q[e^f] \leq D_{\text{KL}}(P||Q)$$

It is clear that if the equality in the Jensen's inequality is attained, then we are done. Note that the function $(t) = -\log(t)$ is strictly convex and finite in $t > 0$ ($\log$ is strictly concave).

By the above corollary, the equality holds iff $\frac{e^{f^*}}{p} = c$ almost surely, where c is the mean of $\frac{e^f}{p}$ and f^* is our optimal function. Then

$$\frac{e^{f^*}}{p} = c \quad e^{f^*} = cp \quad f^* = \log p - \log c = \log \left(\frac{dP}{dQ} \right) - \log c$$

Substitute this optimal function f^* back into the objective:

$$\begin{aligned} & \mathbb{E}_P[f^*] - \log \mathbb{E}_Q[e^{f^*}] \\ &= \mathbb{E}_P[\log(p) - \log(c)] - \log \mathbb{E}_Q[e^{\log(p) - \log(c)}] \\ &= \mathbb{E}_P[\log(p)] - \log(c) - \log(c \cdot \mathbb{E}_Q[p]) \\ &= D_{\text{KL}}(P||Q) - \log(c) - \log(c) - \log(1) \\ &= D_{\text{KL}}(P||Q) \end{aligned}$$

where

$$\mathbb{E}_Q[p] = 1$$

since $\int p dQ = \int dP = 1$ and P is a probability measure, so $\int_{\mathcal{X}} p dQ = \int_{\mathcal{X}} dP = 1$.

Therefore the sup is attained by f^* .

Note that from the derivation above, such f^* is not unique, we can take any valid c we want. We usually take $c = 1$.

□

Proof of Theorem 3.2, first part

Theorem 10.

Let $h_{\text{gum},\Delta} : [0, 1] \rightarrow \mathbb{R}$ be defined as

$$h_{\text{gum},\Delta}(r) = \log \left(\left\lfloor \frac{1}{1-\Delta} \right\rfloor r^{\frac{\Delta}{1-\Delta}} + r^{\frac{\Delta}{1-\Delta}} \right) \quad \text{with} \quad \Delta = (1 - \Delta) \left\lfloor \frac{1}{1-\Delta} \right\rfloor.$$

This function gives the optimal $\mathcal{P}_\Delta$ -efficiency rates in the sense that $\mathcal{P}_\Delta(h_{\text{gum},\Delta}) \geq \mathcal{P}_\Delta(h)$ for any measurable function h . Moreover, $\mathcal{P}_\Delta(h_{\text{gum},\Delta})$ is attained at the following least-favorable NTP distribution in $\mathcal{P}_\Delta$:

$$P_\Delta = \underbrace{1 - \Delta, \dots, 1 - \Delta}_{\left\lfloor \frac{1}{1-\Delta} \right\rfloor \text{ times}}, 1 - (1 - \Delta) \cdot \left\lfloor \frac{1}{1-\Delta} \right\rfloor, 0, \dots$$

The paper proves $\mathcal{P}_\Delta(h_{\text{gum},\Delta}) \geq \mathcal{P}_\Delta(h)$ by doing the following:

Recall from "optimality via minimax optimization", finding the optimal h that gives $\sup_h \mathcal{P}_\Delta(h)$ is reduced to solving $\min_h \max_{P \in \mathcal{P}_\Delta} \mathbb{E}_P[h(Y)] - \log \mathbb{E}_{1,P} e^{-h(Y)}$.

The paper shows

$$\min_h \max_{P \in \mathcal{P}} (h, P) \geq -D(0, 1, P_\Delta)$$

But the paper also shows the min is attained at

$$\max_{P \in \mathcal{P}_\Delta} (h_{\text{gum}, \Delta}, P) = -D(0, 1, P_\Delta)$$

And so this $h_{\text{gum}, \Delta}$ is the (*unique*) score function that solves the minimax problem

The following are the technical details:

We first claim that the radon-nikodym derivative $\frac{d_{1, P_\Delta}}{d_0}$ exists, that is, $1, P_\Delta \ll 0$.

Proof.

It is clear that 0 , which is Y_t^{gum} under H_0 , follows a $U(0, 1)$ distribution. Hence 0 is the Lebesgue measure on $[0, 1]$. Thus for any mble set A , $0(A) = \int_{A[0,1]} 1 dr = 0$ implies $A \cap [0, 1]$ has length 0.

$1, P_\Delta$, which is the distribution of Y_t^{gum} under H_1 at the "least-favorable NTP distribution". By lemma 3.1, we have CDF

$$\sum_{w \in \mathcal{W}} P_w r^{1P_w}$$

for $r \in (0, 1)$. It is clear that since $U_{t,w} \sim U(0, 1)$, for $r \leq 0$ and $r \geq 1$, the CDF is 0 and 1 respectively. Thus we see the support of $1, P_\Delta$ is between $[0, 1]$.

Now

$$1, P_\Delta(A) = \int_{A[0,1]} f_1(r) dr$$

where $f_1(r)$ is the density of $1, P_\Delta$ distribution (we will derive it later).

Therefore, if $0(A) = 0$, then $1, P_\Delta(A) = 0$, hence $1, P_\Delta \ll 0$.

□

Thus we can define the radon-nikodym derivative

$$\frac{d_{1, P_\Delta}}{d_0}$$

Consider equation 21:

To apply the Donsker-Varadhan representation (we are allowed to do so because the radon-nikodym derivative exists), use the substitution $f = -h$:

$$\begin{aligned} \min_h \left(\mathbb{E}_0[h(Y)] - \log \mathbb{E}_{1, P}[e^{-h(Y)}] \right) &= \min_f \left(\mathbb{E}_0[-f(Y)] - \log \mathbb{E}_{1, P}[e^{f(Y)}] \right) \\ &= -\max_f \left(\mathbb{E}_0[f(Y)] - \log \mathbb{E}_{1, P}[e^{f(Y)}] \right) \end{aligned}$$

By the Donsker-Varadhan representation (with $P = \mu_0$ and $Q = \mu_{1,P}$),

$$-\max_f (\mathbb{E}_0[f] - \log \mathbb{E}_{1,P}[e^f]) = -D_{\text{KL}}(\mu_0 \| \mu_{1,P})$$

(It can be proven that $\mu_0 \ll \mu_{1,P}$ just like what we did for $\mu_{1,P_\Delta} \ll \mu_0$, except we need $f_1 > 0$ almost everywhere, indeed $f_1(r) > 0$ for $r \in (0, 1)$)

We know what is this f , it is the f^* that we derived before! But we do not know what is $\log \left(\frac{dP}{dQ} \right)$.

By definition,

$$\mu_0(A) = \int_A f_0(r) \, dr$$

$$\mu_{1,P_\Delta}(A) = \int_A f_1(r) \, dr$$

where f_0, f_1 are the density function of distributions μ_0 and μ_{1,P_Δ} .

We wish to find g such that

$$\mu_{1,P_\Delta}(A) = \int_A g(r) \, d\mu_0$$

From definition, it suffices to find

$$\mu_{1,P_\Delta}(A) = \int_A g(r) (f_0(r) \, dr)$$

Compare with above (because $f_0(r) = 1 \neq 0$),

$$g(r) = \frac{f_1(r)}{f_0(r)}$$

Therefore

$$\frac{d_{1,P_\Delta}}{d_0} = \frac{f_1(r)}{f_0(r)}$$

Now back to the proof of theorem 3.2, in equation 21,

As we have seen in the proof of The Donsker-Varadhan Representation, the specific $h = -f$ that makes the equality hold (which is also our desired minimizer h) is (by taking $c = 1$, so $\log c = 0$)

$$h_{\text{gum},\Delta}(r) = \log \frac{d_{1,P_\Delta}}{d_0}(r) = \log \left(\frac{f_1(r)}{f_0(r)} \right)$$

Now we need to figure out what that is. It is clear that $f_0(r) = 1$

What is $f_1(r)$?

By lemma 3.1, under H_1 , the distribution of Y_t^{gum} given P_t obeys

$$\mathbb{P}_{H_1}(Y_t^{gum} \leq r \mid P_t) = \sum_{w \in \mathcal{W}} P_{t,w} r^{1P_{t,w}} \quad \text{for } r \in [0, 1]$$

For $P_w = 0$, the corresponding term is defined as 0.

By taking its derivative, we obtain the density:

$$\frac{d}{dr} \left(\sum_{w \in \mathcal{W}} P_{t,w} r^{1P_{t,w}} \right) = \sum_{w \in \mathcal{W}} P_w \left(\frac{1}{P_w} r^{\frac{1}{P_w}-1} \right) = \sum_{w \in \mathcal{W}} r^{\frac{1-P_w}{P_w}}$$

By subbing in P_Δ , we can get $f_1(r)$.

Recall

$$\begin{aligned} P_\Delta &= 1 - \Delta, \dots, 1 - \Delta, 1 - (1 - \Delta) \cdot \left\lfloor \frac{1}{1 - \Delta} \right\rfloor, 0, \dots \\ &= 1 - \Delta, \dots, 1 - \Delta, 1 - \Delta, 0, \dots \end{aligned}$$

$\lfloor \frac{1}{1-\Delta} \rfloor$ times

by defining $\Delta = (1 - \Delta) \lfloor \frac{1}{1 - \Delta} \rfloor$.

Therefore

$$f_1(r) = \left\lfloor \frac{1}{1 - \Delta} \right\rfloor r^{\frac{\Delta}{1-\Delta}} r^{\frac{\Delta}{1-\Delta}}$$

and we get the desired $h_{gum,\Delta}$, which is non-decreasing in r .

And the rest is clear by the paper.

□

Proof of lemma 3.2, 3.3 and lemma 3.4

lemma 3.2:

derivative: Use $r^{1P_i} = e^{\frac{\ln r}{P_i}}$

lemma 3.3:

Remark 11.

$\mathcal{P}_\Delta$ is compact and convex by definition

Proof.

By definition $\mathcal{P}_\Delta := \{P : \max_{w \in \mathcal{W}} P_w \leq 1 - \Delta\}$, which is

$$\mathcal{P}_\Delta = \left\{ \mathbf{P} \in \mathbb{R}^{|\mathcal{W}|} : \sum_{w \in \mathcal{W}} P_w = 1, \quad 0 \leq P_w \leq 1 - \Delta \quad \text{for all } w \in \mathcal{W} \right\}$$

Verification of convexity is clear.

To prove compactness, since we are in $\mathbb{R}^n$, by Heine-Borel, it suffices to prove closed and bounded.

Closeness:

- Let $A := \{\mathbf{P} \in \mathbb{R}^{|\mathcal{W}|} : \sum_{w \in \mathcal{W}} P_w = 1\}$. The function $f(\mathbf{P}) = \sum P_w$ is continuous. The set A is the preimage of the closed singleton set $\{1\}$ under f , meaning A is closed.
 - For each $w \in \mathcal{W}$, $w = \{\mathbf{P} \in \mathbb{R}^{|\mathcal{W}|} : P_w \geq 0\}$. The projection function $g_w(\mathbf{P}) = P_w$ is continuous. The set w is the preimage of the closed interval $[0, \infty)$ under g_w , meaning w is closed.
 - For each $w \in \mathcal{W}$, $w = \{\mathbf{P} \in \mathbb{R}^{|\mathcal{W}|} : P_w \leq 1 - \Delta\}$. Using the same continuous projection $g_w(\mathbf{P}) = P_w$, w is the preimage of the closed interval $(-\infty, 1 - \Delta]$, meaning w is closed.
- Note

$$\mathcal{P}_\Delta = A \left(\begin{matrix} w \\ w \in \mathcal{W} \end{matrix} \right) \left(\begin{matrix} w \\ w \in \mathcal{W} \end{matrix} \right)$$

Thus it is closed.

Boundedness:

$$\|\mathbf{P}\|_2 = \sum_{w \in \mathcal{W}} P_w^2 \leq \sum_{w \in \mathcal{W}} 1^2 = |\mathcal{W}|$$

lemma 3.4:

□

What does it mean by **permutation invariance**:

$$h(\mathbf{P}) = \int_0^1 e^{-h(r)} dF_{1,P}(r)$$

where $F_{1,P}(r) = \sum_{w=1}^{|\mathcal{W}|} P_w r^{1P_w}$.

Proof of lemma 3.4:

What does it mean by "It suffices to demonstrate that any point in $\mathcal{P}_\Delta$ is in the convex hull of $(\mathbf{P}_\Delta)$, denoted by $\text{on}((\mathbf{P}_\Delta))$ ":

We wish to prove

$$\mathcal{P}_\Delta = \text{on}((\mathbf{P}_\Delta^*))$$

because of the following:

Remark 12.

If a convex set is generated by the convex hull of a set of points , then the extreme points of that convex set must necessarily belong to this set of points .

“It is easy to find that P is the convex interpolation of $\|\$ points in

(P_Δ) whose first coordinates are the same as $1 - \Delta$

and the i th coordinate is $1 - (1 - \Delta)$ ”：对于集合 中的每一个索引 i ，我们在极值点集合

(P_Δ^*) 中选取一个特定的极值点 i 。这个极值点 i ：前 m 个坐标等于 $1 - \Delta$ ，第 i 个坐标等于剩余概率，其余所有坐标等于 0。为每个 i 分配一个权重 $w_i = \frac{P_i}{1 - \Delta}$ 。对于前 m 个坐标：

$\sum_{i \in} w_i(1 - \Delta) = (1 - \Delta) \sum w_i = 1 - \Delta$ 。与 P 一致。对于第 $j \in$ 个坐标：在所有选取的极值点中，只有 j 在这个位置非零。因此其值为 $w_j \cdot = \frac{P_j}{1 - \Delta} \cdot = P_j$ 。与 P 一致。对于其他坐标：恒为 0，与 P 一致。

In (a), for

$$P = P_1 (1 -)P_2$$

need a valid . On m -th coordinates,

$$P_m = (1 - \Delta) (1 -)(P_m P_{m1} - (1 - \Delta))$$

So

$$= \frac{(1 - \Delta) - P_{m1}}{((1 - \Delta) - P_m) ((1 - \Delta) - P_{m1})}$$

which is in $(0,1)$ as $1 - \Delta > P_m \geq P_{m1}$. Sub in to $m - 1$ coordinates we also get P_{m1} .